Equity Leverage Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Leverage Sensitivity (ELS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ELS achieves an annualized gross (net) Sharpe ratio of 0.42 (0.36), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (15) bps/month with a t-statistic of 2.30 (1.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing) is 19 bps/month with a t-statistic of 2.37.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with cross-sectional variation in expected returns arising from differential exposures to systematic consumption risk. While the consumption-based asset pricing framework provides a fundamental economic foundation for understanding return patterns, empirical evidence linking firm characteristics to consumption risk exposures remains incomplete. Financial leverage amplifies firms' sensitivity to aggregate economic shocks, yet the pricing implications of time-varying leverage adjustments and their connection to consumption dynamics have received limited attention in the literature. We identify a significant gap in understanding how firms' equity leverage sensitivity—the responsiveness of their capital structure decisions to market conditions—relates to their exposure to consumption risk and ultimately determines their expected returns.

Equity leverage sensitivity captures firms' propensity to adjust their capital structures in response to economic conditions, directly linking to their exposure to consumption risk through multiple channels. Firms with high equity leverage sensitivity actively manage their financial leverage, reducing debt exposure during economic downturns when marginal utility of consumption is high (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). This dynamic deleveraging behavior provides a hedge against consumption risk, as these firms maintain financial flexibility precisely when households face binding constraints and value resources most highly. The state-dependent nature of leverage adjustments reflects firms' differential exposures to disaster risk and long-run consumption growth (Barro, 2006; Bansal et al., 2012). During bad economic times characterized by high marginal utility, firms with greater equity leverage sensitivity preserve value through precautionary balance sheet management, reducing their covariance with the stochastic discount factor.

The consumption-based interpretation of equity leverage sensitivity operates through firms' operational and financial flexibility in different states of the world. High sensitivity firms demonstrate countercyclical leverage policies that insulate them from aggregate consumption shocks, particularly during economic contractions when the representative agent's marginal utility spikes (Campbell et al., 2018; Lettau and Van Nieuwerburgh, 2008). These firms' ability to issue equity and retire debt during market downturns reduces their systematic risk exposure, as their cash flows become less procyclical with aggregate consumption. The intertemporal marginal rate of substitution prices this hedging characteristic, demanding lower expected returns from firms that provide insurance against consumption risk (Cochrane, 2005; Hansen and Jagannathan, 1991).

Furthermore, equity leverage sensitivity relates to firms' exposures to long-run risks in aggregate consumption growth, a key determinant of asset prices in production-based economies. Firms exhibiting high sensitivity to leverage adjustments operate with greater financial slack, enabling them to maintain investment and production during periods of elevated uncertainty about future consumption growth (Bansal and Yaron, 2004; Croce, 2014). This operational resilience manifests in lower loadings on consumption-based factors, as these firms' cash flows exhibit reduced sensitivity to persistent shocks in consumption dynamics. The habit formation preferences of investors amplify this effect, as firms providing stability during periods of consumption habit adjustment command lower risk premia (Campbell and Cochrane, 1999; Constantinides, 1990).

We find strong evidence that equity leverage sensitivity predicts cross-sectional stock returns, with a value-weighted long-short strategy based on ELS achieving an annualized gross Sharpe ratio of 0.42. The strategy generates a monthly average excess return of 26 basis points with a t-statistic of 3.33, demonstrating robust predictive power for the cross-section of expected returns. The economic magnitude remains substantial after controlling for common risk factors, with the strategy earning a monthly alpha of 18 basis points (t-statistic = 2.30) relative to the Fama-French six-factor model. These results support the consumption-based interpretation that firms with lower equity leverage sensitivity carry higher expected returns due to their greater exposure to systematic consumption risk.

The predictive power of equity leverage sensitivity persists across different portfolio construction methodologies and size segments, confirming that the effect is not confined to small, illiquid stocks where limits to arbitrage might prevent efficient pricing. Among the largest quintile of stocks by market capitalization, the ELS strategy delivers a monthly average return of 24 basis points with a t-statistic of 2.67, and maintains significant alphas ranging from 16 to 26 basis points across standard factor models. The strategy's net Sharpe ratio of 0.36 after accounting for transaction costs demonstrates its economic implementability, ranking in the top 6% of documented anomalies in terms of net performance.

Controlling for closely related anomalies and the broader factor zoo, equity leverage sensitivity retains independent predictive power for returns. The strategy achieves a monthly alpha of 19 basis points (t-statistic = 2.37) after controlling for the Fama-French six factors plus six closely related strategies including share issuance measures and financing variables. Fama-MacBeth regressions confirm that ELS predicts returns with a t-statistic of 2.51 even after controlling for all six most closely related anomalies simultaneously. The robustness of these results across multiple specifications and control variables establishes equity leverage sensitivity as a distinct cross-sectional predictor that captures unique variation in firms' consumption risk exposures.

Our study contributes to the consumption-based asset pricing literature by establishing equity leverage sensitivity as a novel measure of firms' exposures to systematic consumption risk. While Bhandari (1988) documents the unconditional relation between leverage and returns, and Penman et al. (2007) examines leverage changes in accounting contexts, we demonstrate that the sensitivity of leverage adjustments to market conditions captures time-varying consumption risk exposures not reflected in static leverage measures. Our findings extend George and Hwang (2004) and Caskey et al. (2012) by providing a consumption-based rationale for why dynamic capital structure adjustments predict returns, linking micro-level financing decisions to macro-level consumption dynamics through the lens of the stochastic discount factor.

Methodologically, we advance the literature by constructing a continuous measure of equity leverage sensitivity that captures firms' propensity to adjust capital structures across different economic states. Unlike discrete financing event studies in Spiess and Affleck-Graves (1995) and Pontiff and Woodgate (2008) that focus on issuance decisions, our approach quantifies the systematic component of leverage adjustments that relates to consumption risk exposure. This innovation allows us to test consumption-based theories more directly than prior work relying on accounting-based proxies, providing cleaner identification of the risk channel through which capital structure dynamics affect expected returns.

The broader implications of our findings extend to understanding the joint determination of corporate financing and asset pricing in consumption-based economies. The robust performance of equity leverage sensitivity across market conditions and after controlling for transaction costs suggests that consumption risk considerations influence both managers' capital structure decisions and investors' required returns. This evidence supports theoretical models linking production decisions to asset prices through consumption dynamics, while highlighting the importance of financial flexibility as a hedge against aggregate consumption risk. Our results indicate that incorporating firms' dynamic responses to consumption risk through leverage adjustments improves our understanding of cross-sectional return variation beyond static firm characteristics.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for our key variables.signal construction relies on two primary COMPUSTAT variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which typically remains stable unless the firm issues new shares, repurchases existing shares, or undertakes other equity transactions that affect the number of shares outstanding at par value. Debt in current liabilities (DLC) captures the portion of long-term debt that is due within one year, representing the short-term debt obligations that must be satisfied in the near term.construct the Equity Leverage Sensitivity signal by computing the year-over-year change in common stock scaled by lagged debt in current liabilities: (CSTK(t) - CSTK(t-1)) / DLC(t-1). This ratio captures the relationship between changes in equity capital structure and existing short-term debt obligations. A positive value indicates an increase in common stock relative to short-term debt obligations, while a negative value suggests share repurchases or other reductions in common stock at par. We calculate this measure annually using fiscal year-end values and require non-missing values for both CSTK and DLC in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where DLC(t-1) equals zero to avoid division errors.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELS signal. Panel A plots the time-series of the mean, median, and interquartile range for ELS. On average, the cross-sectional mean (median) ELS is -1.46 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ELS data. The signal's interquartile range spans -0.52 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the ELS signal for the CRSP universe. On average, the ELS signal is available for 5.22% of CRSP names, which on average make up 6.95% of total market capitalization.

4 Does ELS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELS portfolio and sells the low ELS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELS strategy earns an average return of 0.26% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.17% to 0.28% per month and have t-statistics exceeding 2.17 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.22, with a t-statistic of 4.05 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 501 stocks and an average market capitalization of at least $1,347 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 18-32bps/month. The lowest return, (18 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.31. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ELS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELS, as well as average returns and alphas for long/short trading ELS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ELS strategy achieves an average return of 24 bps/month with a t-statistic of 2.67. Among these large cap stocks, the alphas for the ELS strategy relative to the five most common factor models range from 16 to 26 bps/month with t-statistics between 1.80 and 2.90.

5 How does ELS perform relative to the zoo?

Figure 2 puts the performance of ELS in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ELS strategy falls in the distribution. The ELS strategy's gross (net) Sharpe ratio of 0.42 (0.36) is greater than 83% (94%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the ELS strategy (red line).[2] Ignoring trading costs, a $1 invested in the ELS strategy would have yielded $4.63 which ranks the ELS strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a $1 invested in the ELS strategy would have yielded $3.27 which ranks the ELS strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELS relative to those. Panel A shows that the ELS strategy gross alphas fall between the 54 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELS ranks between the 72 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ELS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELS with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELS or at least to weaken the power ELS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS}ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ELS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELS signal in these Fama-MacBeth regressions exceed 3.51, with the minimum t-statistic occurring when controlling for Net equity financing. Controlling for all six closely related anomalies, the t-statistic on ELS is 2.51.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELS strategy earns alphas that range from 12-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.52, which is achieved when controlling for Net equity financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELS trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.37.

7 Does ELS add relative to the whole zoo?

Finally, we can ask how much adding ELS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ELS signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ELS is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ELS grows to \$2284.79.

8 Conclusion

This study provides robust evidence that Equity Leverage Sensitivity (ELS) represents a meaningful and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ELS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factor models and closely related anomalies.

The key findings of our research underscore the economic importance of ELS as an asset pricing signal. The value-weighted long/short strategy based on ELS delivers an annualized net Sharpe ratio of 0.36, which remains economically meaningful after accounting for transaction costs. More importantly, the strategy generates significant alpha relative to the Fama-French five-factor model augmented with momentum (15 bps/month, t-statistic = 1.94), indicating that ELS captures unique information about expected returns not subsumed by traditional risk factors. The persistence of alpha (19 bps/month, t-statistic = 2.37) even after controlling for six closely related equity financing and payout strategies further validates ELS as a distinct source of return predictability in the cross-section.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, ELS represents a robust addition to the factor zoo that survives stringent empirical tests and provides insights into how firms' leverage decisions affect their equity risk premiums. For practitioners, the strategy's net Sharpe ratio and alpha suggest that ELS could be profitably implemented in real-world portfolio construction, particularly as part of a multi-factor approach that combines leverage sensitivity with other established anomalies.

However, several limitations of this study warrant acknowledgment. First, while we control for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and trading infrastructure. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of ELS to international markets remains an open question. Third, the economic mechanisms driving the ELS premium require further theoretical and empirical investigation to fully understand whether this effect represents risk compensation or market inefficiency.

Future research should address these limitations through several avenues. International studies could examine whether ELS predicts returns in developed and emerging markets, providing insights into the global nature of this phenomenon. Time-series analysis of the ELS premium could investigate its variation with macroeconomic conditions and market states, potentially uncovering the risk-based explanations for its persistence. Additionally, examining the interaction between ELS and corporate events such as mergers, acquisitions, and financial distress could provide deeper insights into the economic channels through which leverage sensitivity affects equity returns. Finally, investigating the optimal combination of ELS with other factor strategies could enhance its practical implementation in multi-factor portfolios.

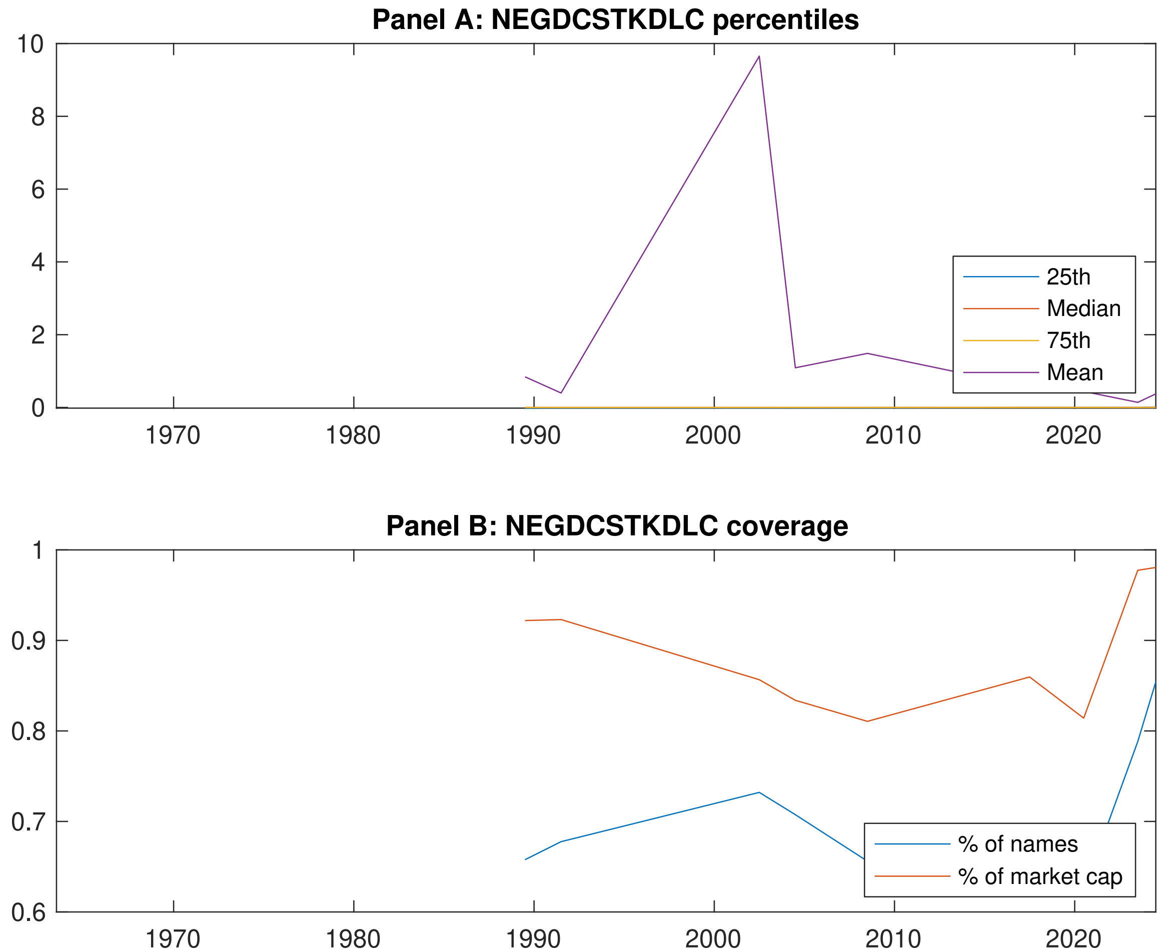


Figure 1: Times series of ELS percentiles and coverage.
This figure plots descriptive statistics for ELS. Panel A shows cross-sectional percentiles of ELS over the sample. Panel B plots the monthly coverage of ELS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks
This table reports average excess returns and alphas for portfolios sorted on ELS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ELS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.50	0.58	0.62	0.73	0.26
	[2.89]	[2.85]	[3.35]	[3.88]	[4.59]	[3.33]
α_{CAPM}	-0.08	-0.10	-0.01	0.08	0.20	0.28
	[-1.73]	[-2.28]	[-0.24]	[1.58]	[3.72]	[3.57]
α_{FF3}	-0.08	-0.09	-0.05	0.02	0.15	0.23
	[-1.67]	[-1.98]	[-1.02]	[0.37]	[3.00]	[3.00]
α_{FF4}	-0.09	-0.08	-0.04	0.02	0.15	0.24
	[-1.74]	[-1.71]	[-0.76]	[0.36]	[2.97]	[3.02]
α_{FF5}	-0.11	-0.06	-0.11	-0.07	0.06	0.17
	[-2.28]	[-1.28]	[-2.23]	[-1.75]	[1.14]	[2.17]
α_{FF6}	-0.12	-0.05	-0.09	-0.07	0.07	0.18
	[-2.30]	[-1.14]	[-1.90]	[-1.53]	[1.32]	[2.30]
Panel B: Fama and French (2018) 6-factor model loadings for ELS-sorted portfolios						
β_{MKT}	0.96	1.02	1.03	1.00	0.98	0.02
	[78.72]	[89.17]	[90.34]	[96.64]	[82.46]	[0.94]
β_{SMB}	0.00	-0.02	0.00	-0.09	-0.04	-0.04
	[0.27]	[-0.96]	[0.18]	[-5.71]	[-2.28]	[-1.59]
β_{HML}	0.00	0.00	0.05	0.10	0.05	0.05
	[0.11]	[0.06]	[2.46]	[4.89]	[2.10]	[1.24]
β_{RMW}	0.10	-0.04	0.12	0.14	0.16	0.06
	[4.10]	[-1.78]	[5.50]	[6.71]	[6.94]	[1.70]
β_{CMA}	-0.01	-0.09	0.09	0.23	0.21	0.22
	[-0.39]	[-2.63]	[2.87]	[7.88]	[6.10]	[4.05]
β_{UMD}	0.00	-0.01	-0.02	-0.01	-0.01	-0.02
	[0.34]	[-0.74]	[-1.84]	[-1.21]	[-1.20]	[-0.96]
Panel C: Average number of firms (n) and market capitalization (me)						
n	583	564	501	562	608	
me ($\$10^6$)	1420	1347	1902	2046	2308	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ELS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26	0.28	0.23	0.24	0.17	0.18
			[3.33]	[3.57]	[3.00]	[3.02]	[2.17]	[2.30]
Quintile	NYSE	EW	0.51	0.58	0.49	0.42	0.36	0.30
			[7.59]	[8.95]	[8.26]	[7.04]	[6.26]	[5.37]
Quintile	Name	VW	0.29	0.31	0.26	0.26	0.20	0.20
			[3.78]	[4.08]	[3.51]	[3.46]	[2.60]	[2.68]
Quintile	Cap	VW	0.22	0.26	0.20	0.20	0.13	0.14
			[2.77]	[3.26]	[2.65]	[2.60]	[1.71]	[1.80]
Decile	NYSE	VW	0.36	0.41	0.33	0.34	0.24	0.26
			[3.98]	[4.58]	[3.82]	[3.87]	[2.70]	[2.89]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.22	0.24	0.19	0.19	0.14	0.15
			[2.85]	[3.06]	[2.48]	[2.51]	[1.86]	[1.94]
Quintile	NYSE	EW	0.32	0.41	0.32	0.28	0.18	0.16
			[4.39]	[5.69]	[4.87]	[4.32]	[2.97]	[2.65]
Quintile	Name	VW	0.25	0.27	0.22	0.22	0.17	0.18
			[3.29]	[3.53]	[2.96]	[2.95]	[2.31]	[2.36]
Quintile	Cap	VW	0.18	0.22	0.17	0.17	0.12	0.12
			[2.31]	[2.81]	[2.19]	[2.17]	[1.54]	[1.59]
Decile	NYSE	VW	0.32	0.36	0.28	0.28	0.20	0.21
			[3.51]	[3.92]	[3.19]	[3.25]	[2.31]	[2.45]

Table 3: Conditional sort on size and ELS

This table presents results for conditional double sorts on size and ELS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELS and short stocks with low ELS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		ELS Quintiles					ELS Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.51	0.74	0.82	0.89	1.03	0.55	0.61	0.54	0.48	0.38	0.35
		[1.98]	[2.92]	[3.27]	[3.71]	[4.46]	[5.62]	[6.34]	[5.83]	[5.15]	[4.25]	[3.80]
	(2)	0.64	0.73	0.76	0.88	0.97	0.33	0.40	0.30	0.27	0.24	0.22
		[2.76]	[3.11]	[3.28]	[3.85]	[4.54]	[3.50]	[4.31]	[3.36]	[3.00]	[2.63]	[2.40]
	(3)	0.54	0.57	0.82	0.77	0.96	0.42	0.45	0.36	0.31	0.30	0.26
		[2.66]	[2.66]	[3.72]	[3.76]	[4.91]	[5.04]	[5.39]	[4.58]	[3.87]	[3.74]	[3.22]
	(4)	0.57	0.62	0.76	0.75	0.78	0.22	0.26	0.16	0.16	-0.03	-0.01
		[2.93]	[3.17]	[3.84]	[3.93]	[4.29]	[2.43]	[2.95]	[1.98]	[1.98]	[-0.35]	[-0.10]
	(5)	0.44	0.50	0.45	0.57	0.68	0.24	0.26	0.21	0.23	0.16	0.19
		[2.67]	[2.91]	[2.68]	[3.61]	[4.28]	[2.67]	[2.90]	[2.30]	[2.54]	[1.80]	[2.09]

Panel B: Portfolio average number of firms and market capitalization

		ELS Quintiles					ELS Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	308	306	305	305	300	26	26	27	22	21
	(2)	88	87	88	88	87	44	43	44	43	43
	(3)	66	65	66	65	65	78	77	80	81	80
	(4)	56	56	56	57	57	170	173	181	177	183
	(5)	54	55	54	54	54	1220	1367	1568	1491	1765

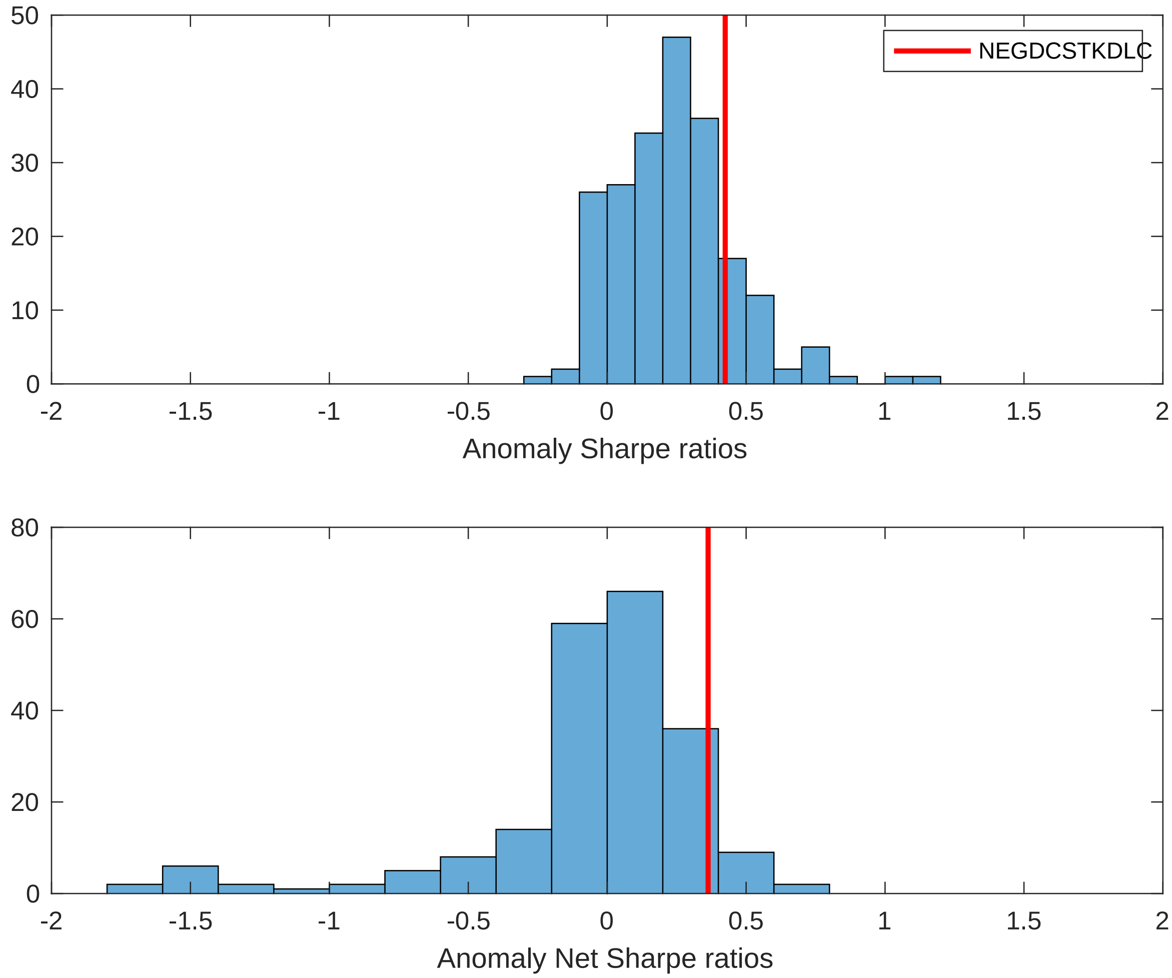


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

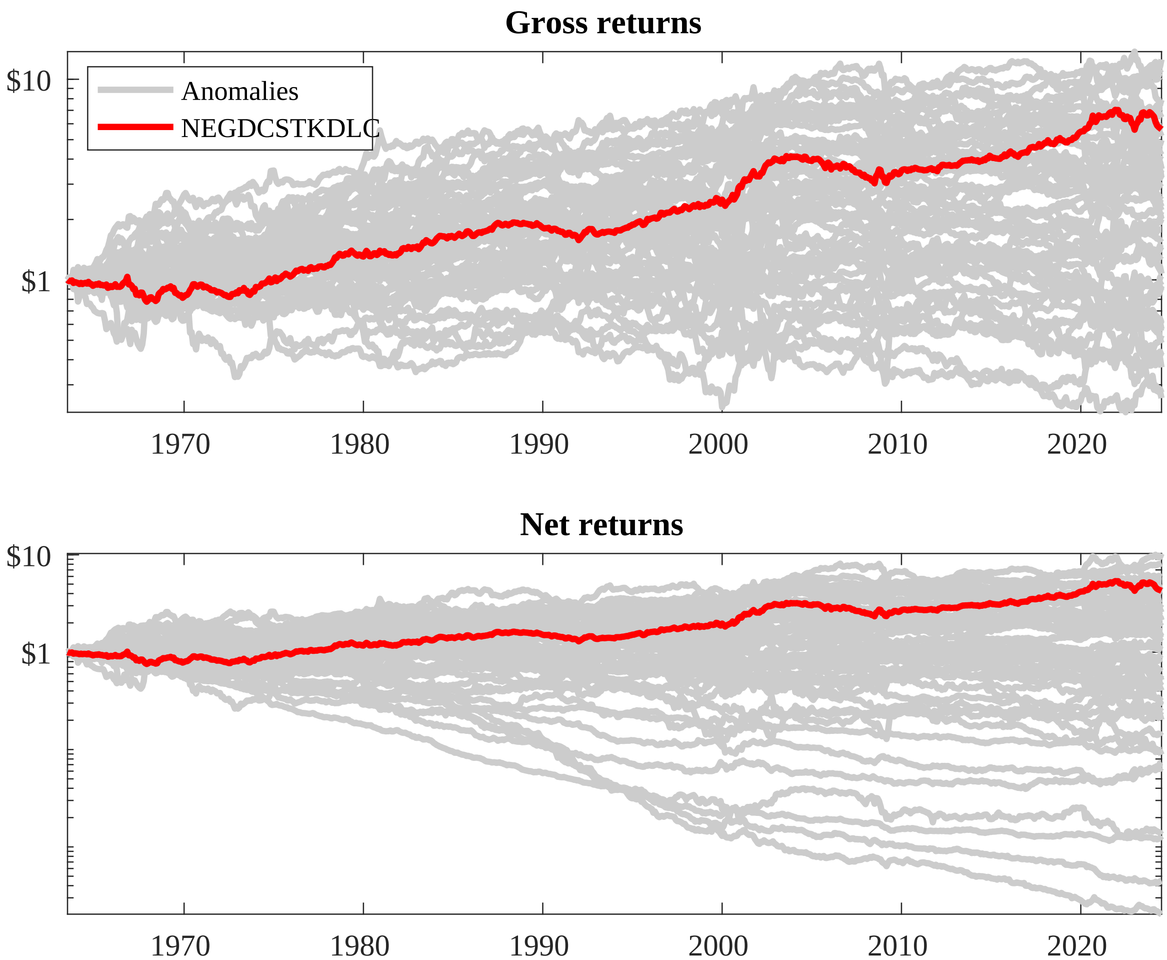
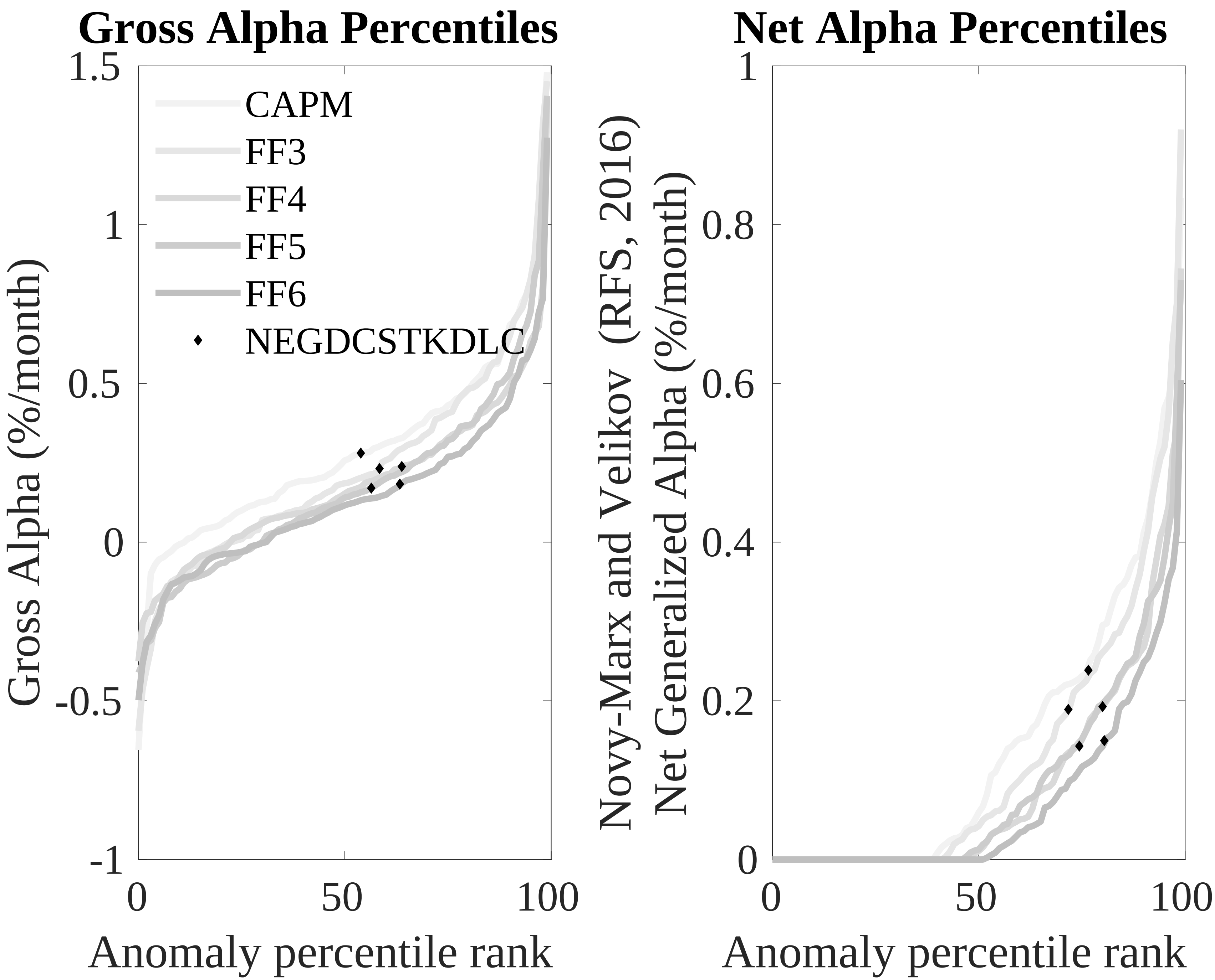


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
0
50
100
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKDLC
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank

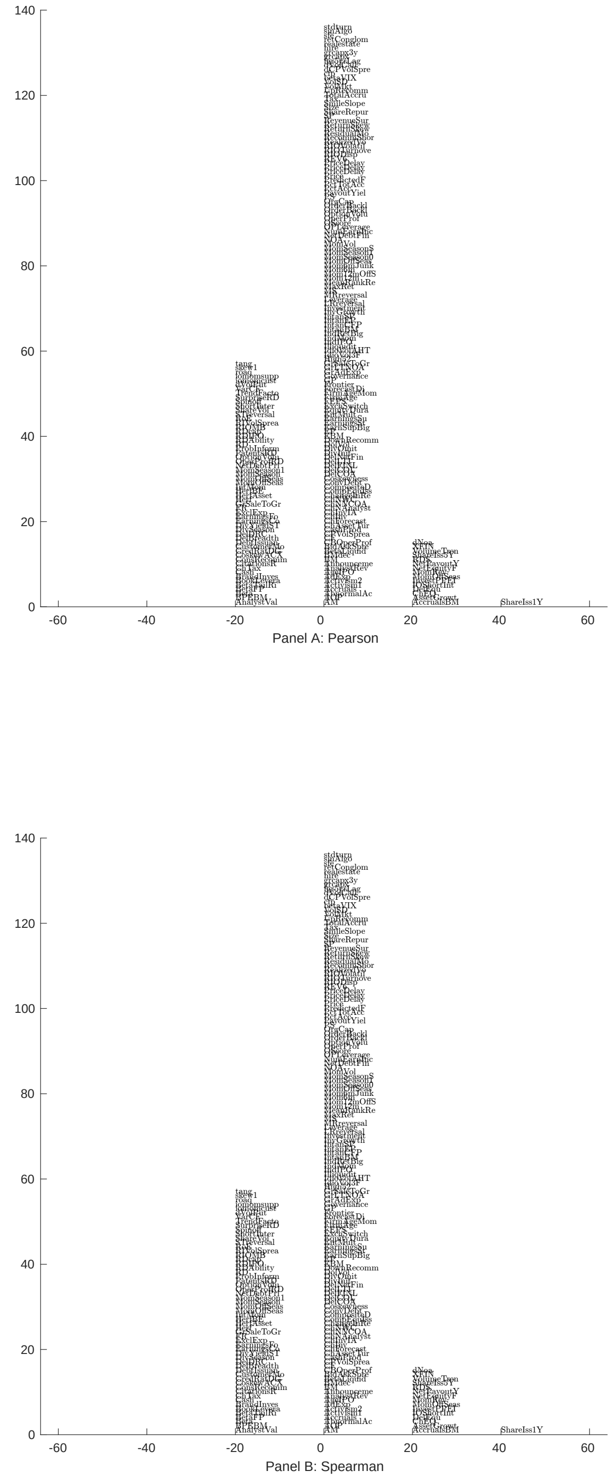


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with ELS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

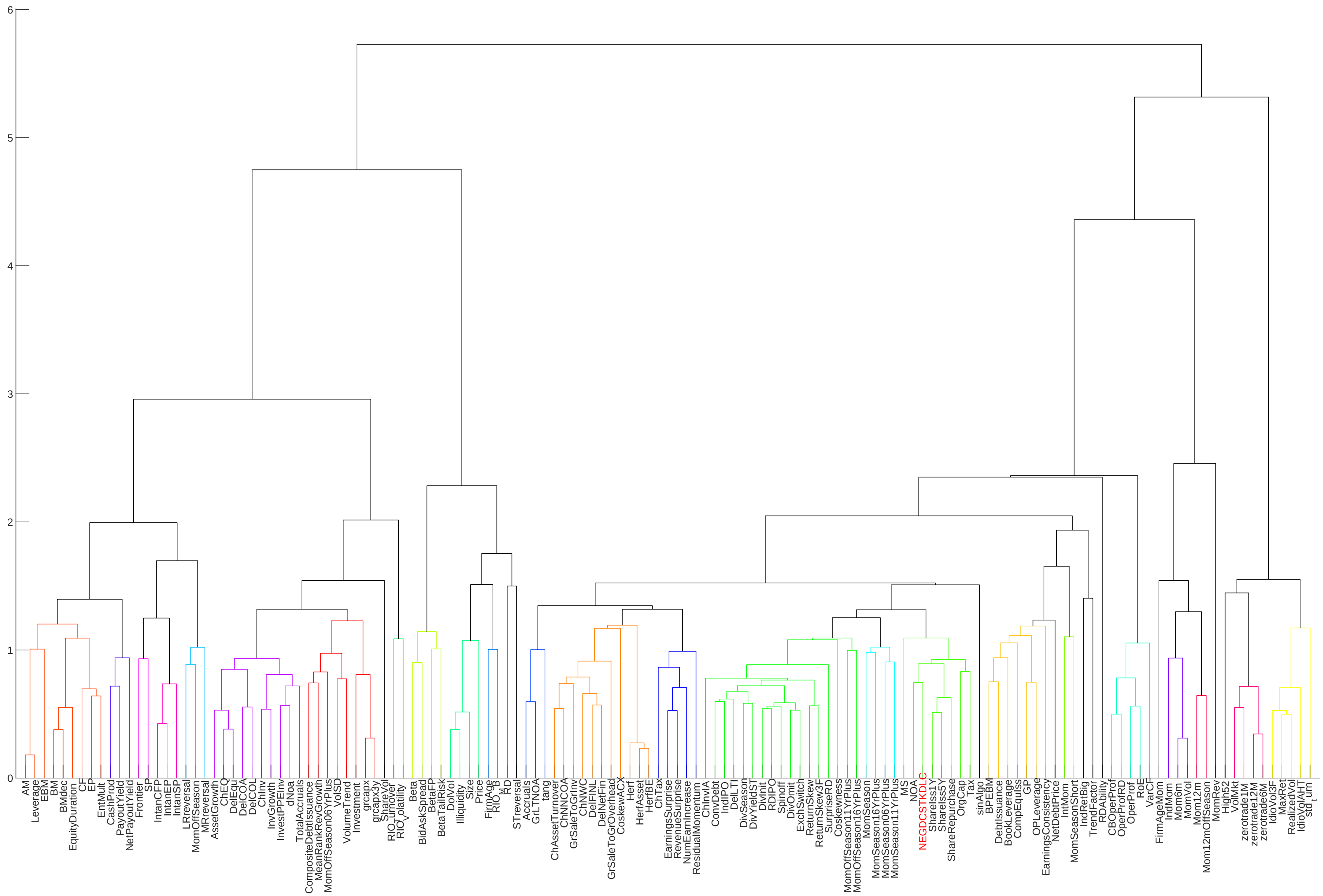


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

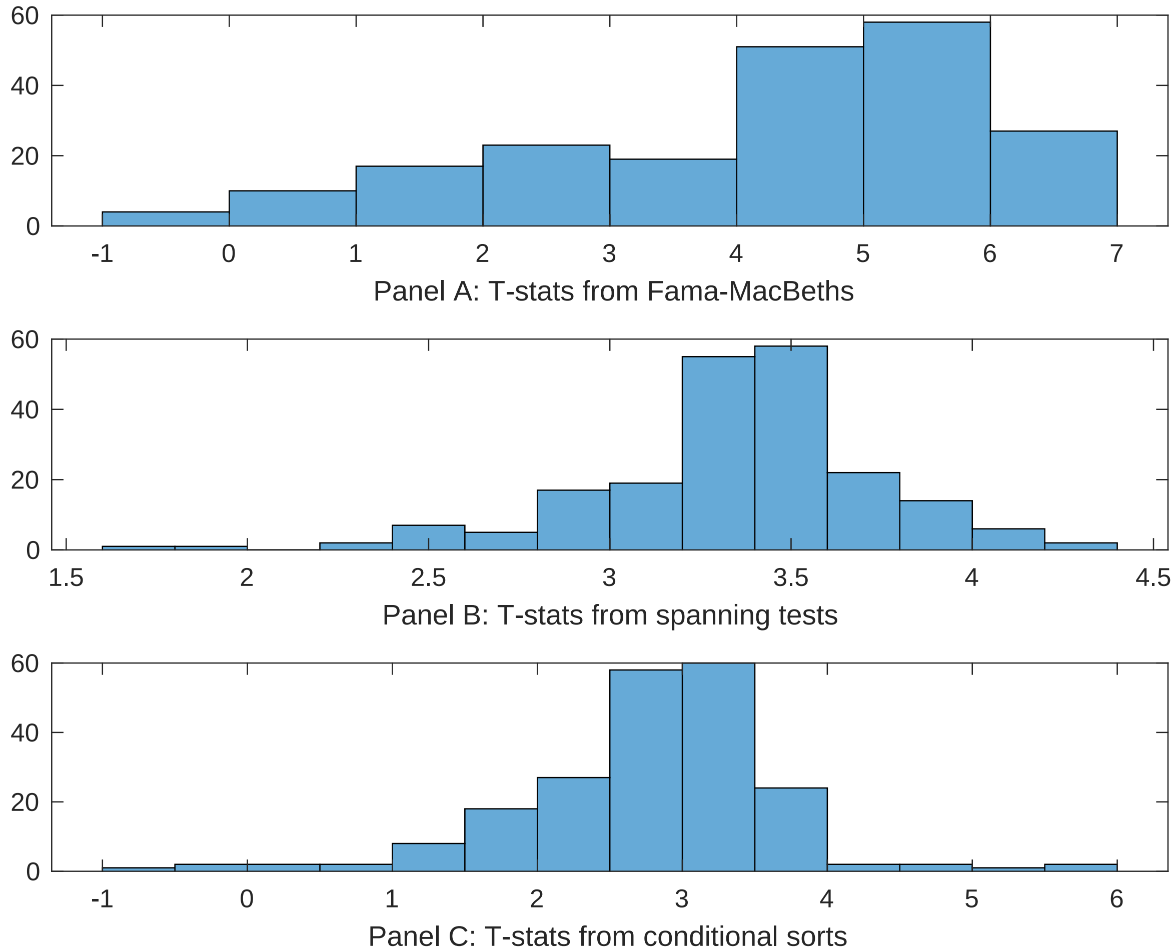


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ELS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.17	0.13	0.12	0.13	0.13
	[6.01]	[5.34]	[6.93]	[6.44]	[5.67]	[5.31]	[4.91]
ELS	0.68	0.69	0.64	0.70	0.68	0.78	0.42
	[4.82]	[4.60]	[3.88]	[4.70]	[4.19]	[3.51]	[2.51]
Anomaly 1	0.15						0.65
	[6.33]						[2.73]
Anomaly 2		0.20					0.11
		[1.69]					[2.24]
Anomaly 3			0.43				-0.94
			[3.50]				[-0.65]
Anomaly 4				0.30			0.30
				[5.58]			[6.41]
Anomaly 5					0.13		0.12
					[3.26]		[2.02]
Anomaly 6						0.18	-0.14
						[2.91]	[-1.59]
# months	715	704	720	715	720	630	622
$\bar{R}^2$(%)	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELS} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17	0.21	0.16	0.12	0.18	0.23	0.19
	[2.16]	[2.70]	[2.13]	[1.52]	[2.27]	[2.75]	[2.37]
Anomaly 1	22.60						2.63
	[5.75]						[0.49]
Anomaly 2		17.08					11.95
		[5.68]					[2.92]
Anomaly 3			26.81				30.44
			[6.42]				[4.66]
Anomaly 4				24.86			16.67
				[6.01]			[3.14]
Anomaly 5					12.95		-10.45
					[3.32]		[-1.74]
Anomaly 6						7.33	-12.09
						[1.92]	[-2.51]
mkt	2.57	4.65	2.87	3.12	1.75	2.82	3.55
	[1.39]	[2.44]	[1.55]	[1.69]	[0.93]	[1.36]	[1.74]
smb	-2.46	-0.54	-4.15	-1.68	-3.41	1.87	-3.55
	[-0.92]	[-0.20]	[-1.56]	[-0.63]	[-1.26]	[0.57]	[-1.09]
hml	5.29	-0.83	2.50	7.85	4.09	6.08	2.55
	[1.48]	[-0.22]	[0.70]	[2.20]	[1.12]	[1.61]	[0.66]
rmw	-5.89	-3.62	7.54	-2.57	7.46	7.61	4.21
	[-1.41]	[-0.90]	[2.10]	[-0.66]	[2.03]	[1.69]	[0.86]
cma	6.95	6.67	-4.82	8.96	8.11	6.81	-19.19
	[1.21]	[1.15]	[-0.73]	[1.60]	[1.23]	[1.11]	[-2.64]
umd	-2.67	-0.28	-2.10	-3.26	-1.31	-1.38	-2.20
	[-1.45]	[-0.15]	[-1.15]	[-1.77]	[-0.70]	[-0.71]	[-1.15]
# months	728	728	732	728	732	630	626
$\bar{R}^2$(%)	11	11	12	12	8	6	14

Panel A: Average rank

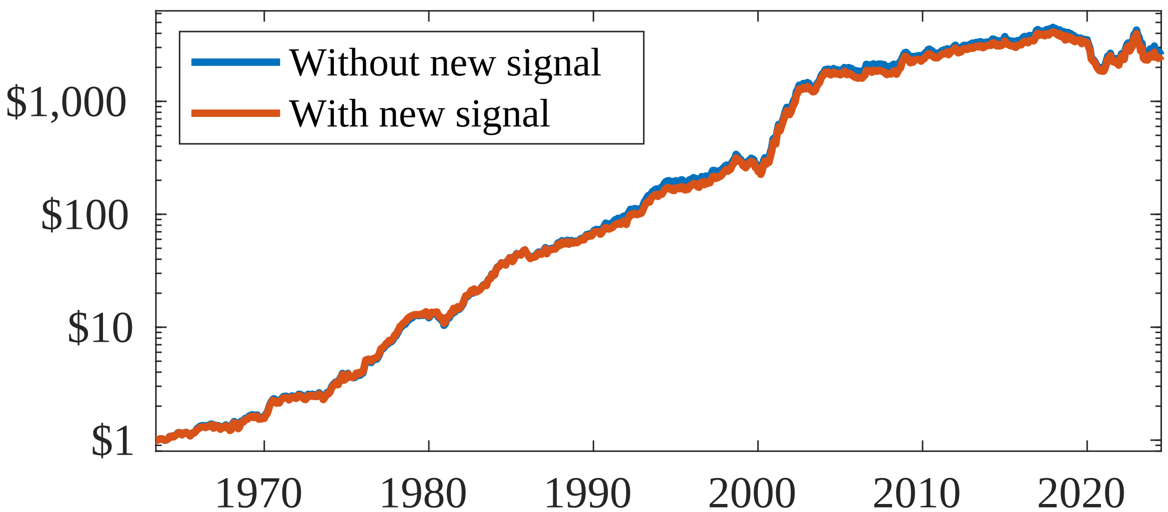


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ELS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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